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LAB ASSIGNMENT - 07

Que1	Run a few randomly-generated problems with just two jobs and two queues; compute the MLFQ execution trace for each. Make your life easier by limiting the length of each job and turning off I/Os.

```
jitanjali_patel@jitanjali:~$ cd os
jitanjali_patel@jitanjali:~/os$ nano 7.py
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 2 -n 2 -q 5 -a 1 -m 20 -M 0 -s 1 -c
Here is the list of inputs:
OPTIONS jobs 2
OPTIONS queues 2
OPTIONS allotments for queue  1 is   1
OPTIONS quantum length for queue  1 is   5
OPTIONS allotments for queue  0 is   1
OPTIONS quantum length for queue  0 is   5
OPTIONS boost 0
OPTIONS ioTime 5
OPTIONS stayAfterIO False
OPTIONS iobump False


For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq    : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job  0: startTime   0 - runTime   3 - ioFreq   0
  Job  1: startTime   0 - runTime  15 - ioFreq   0


Execution Trace:

[ time 0 ] JOB BEGINS by JOB 0
[ time 0 ] JOB BEGINS by JOB 1
[ time 0 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 2 (of 3) ]
[ time 1 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 1 (of 3) ]
[ time 2 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 0 (of 3) ]
```

```
[ time 2 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 0 (of 3) ]
[ time 3 ] FINISHED JOB 0
[ time 3 ] Run JOB 1 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 14 (of 15) ]
[ time 4 ] Run JOB 1 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 13 (of 15) ]
[ time 5 ] Run JOB 1 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 12 (of 15) ]
[ time 6 ] Run JOB 1 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 11 (of 15) ]
[ time 7 ] Run JOB 1 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 10 (of 15) ]
[ time 8 ] Run JOB 1 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 9 (of 15) ]
[ time 9 ] Run JOB 1 at PRIORITY 0 [ TICKS 3 ALLOT 1 TIME 8 (of 15) ]
[ time 10 ] Run JOB 1 at PRIORITY 0 [ TICKS 2 ALLOT 1 TIME 7 (of 15) ]
[ time 11 ] Run JOB 1 at PRIORITY 0 [ TICKS 1 ALLOT 1 TIME 6 (of 15) ]
[ time 12 ] Run JOB 1 at PRIORITY 0 [ TICKS 0 ALLOT 1 TIME 5 (of 15) ]
[ time 13 ] Run JOB 1 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 4 (of 15) ]
[ time 14 ] Run JOB 1 at PRIORITY 0 [ TICKS 3 ALLOT 1 TIME 3 (of 15) ]
[ time 15 ] Run JOB 1 at PRIORITY 0 [ TICKS 2 ALLOT 1 TIME 2 (of 15) ]
[ time 16 ] Run JOB 1 at PRIORITY 0 [ TICKS 1 ALLOT 1 TIME 1 (of 15) ]
[ time 17 ] Run JOB 1 at PRIORITY 0 [ TICKS 0 ALLOT 1 TIME 0 (of 15) ]
[ time 18 ] FINISHED JOB 1

Final statistics:
  Job  0: startTime   0 - response   0 - turnaround   3
  Job  1: startTime   0 - response   3 - turnaround  18

  Avg  1: startTime n/a - response 1.50 - turnaround 10.50
```

Que2 How would you run the scheduler to reproduce each of the examples in the chapter?

```
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 3 -Q 5,10,20 -A 1,2,4 -m 30 -M 0 -s 2 -c
Here is the list of inputs:
OPTIONS jobs 3
OPTIONS queues 3
OPTIONS allotments for queue  2 is   1
OPTIONS quantum length for queue  2 is   5
OPTIONS allotments for queue  1 is   2
OPTIONS quantum length for queue  1 is  10
OPTIONS allotments for queue  0 is   4
OPTIONS quantum length for queue  0 is  20
OPTIONS boost 0
OPTIONS ioTime 5
OPTIONS stayAfterIO False
OPTIONS iobump False


For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq    : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job  0: startTime   0 - runTime  28 - ioFreq   0
  Job  1: startTime   0 - runTime   2 - ioFreq   0
  Job  2: startTime   0 - runTime  25 - ioFreq   0


Execution Trace:

[ time 0 ] JOB BEGINS by JOB 0
[ time 0 ] JOB BEGINS by JOB 1
[ time 0 ] JOB BEGINS by JOB 2
```

```
[ time 0 ] Run JOB 0 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 27 (of 28) ]
[ time 1 ] Run JOB 0 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 26 (of 28) ]
[ time 2 ] Run JOB 0 at PRIORITY 2 [ TICKS 2 ALLOT 1 TIME 25 (of 28) ]
[ time 3 ] Run JOB 0 at PRIORITY 2 [ TICKS 1 ALLOT 1 TIME 24 (of 28) ]
[ time 4 ] Run JOB 0 at PRIORITY 2 [ TICKS 0 ALLOT 1 TIME 23 (of 28) ]
[ time 5 ] Run JOB 1 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 1 (of 2) ]
[ time 6 ] Run JOB 1 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 0 (of 2) ]
[ time 7 ] FINISHED JOB 1
[ time 7 ] Run JOB 2 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 24 (of 25) ]
[ time 8 ] Run JOB 2 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 23 (of 25) ]
[ time 9 ] Run JOB 2 at PRIORITY 2 [ TICKS 2 ALLOT 1 TIME 22 (of 25) ]
[ time 10 ] Run JOB 2 at PRIORITY 2 [ TICKS 1 ALLOT 1 TIME 21 (of 25) ]
[ time 11 ] Run JOB 2 at PRIORITY 2 [ TICKS 0 ALLOT 1 TIME 20 (of 25) ]
[ time 12 ] Run JOB 0 at PRIORITY 1 [ TICKS 9 ALLOT 2 TIME 22 (of 28) ]
[ time 13 ] Run JOB 0 at PRIORITY 1 [ TICKS 8 ALLOT 2 TIME 21 (of 28) ]
[ time 14 ] Run JOB 0 at PRIORITY 1 [ TICKS 7 ALLOT 2 TIME 20 (of 28) ]
[ time 15 ] Run JOB 0 at PRIORITY 1 [ TICKS 6 ALLOT 2 TIME 19 (of 28) ]
[ time 16 ] Run JOB 0 at PRIORITY 1 [ TICKS 5 ALLOT 2 TIME 18 (of 28) ]
[ time 17 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 2 TIME 17 (of 28) ]
[ time 18 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 2 TIME 16 (of 28) ]
[ time 19 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 2 TIME 15 (of 28) ]
[ time 20 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 2 TIME 14 (of 28) ]
[ time 21 ] Run JOB 0 at PRIORITY 1 [ TICKS 0 ALLOT 2 TIME 13 (of 28) ]
[ time 22 ] Run JOB 2 at PRIORITY 1 [ TICKS 9 ALLOT 2 TIME 19 (of 25) ]
[ time 23 ] Run JOB 2 at PRIORITY 1 [ TICKS 8 ALLOT 2 TIME 18 (of 25) ]
[ time 24 ] Run JOB 2 at PRIORITY 1 [ TICKS 7 ALLOT 2 TIME 17 (of 25) ]
[ time 25 ] Run JOB 2 at PRIORITY 1 [ TICKS 6 ALLOT 2 TIME 16 (of 25) ]
[ time 26 ] Run JOB 2 at PRIORITY 1 [ TICKS 5 ALLOT 2 TIME 15 (of 25) ]
[ time 27 ] Run JOB 2 at PRIORITY 1 [ TICKS 4 ALLOT 2 TIME 14 (of 25) ]
[ time 28 ] Run JOB 2 at PRIORITY 1 [ TICKS 3 ALLOT 2 TIME 13 (of 25) ]
[ time 29 ] Run JOB 2 at PRIORITY 1 [ TICKS 2 ALLOT 2 TIME 12 (of 25) ]
[ time 30 ] Run JOB 2 at PRIORITY 1 [ TICKS 1 ALLOT 2 TIME 11 (of 25) ]
```

```
[ time 31 ] Run JOB 2 at PRIORITY 1 [ TICKS 0 ALLOT 2 TIME 10 (of 25) ]
[ time 32 ] Run JOB 0 at PRIORITY 1 [ TICKS 9 ALLOT 1 TIME 12 (of 28) ]
[ time 33 ] Run JOB 0 at PRIORITY 1 [ TICKS 8 ALLOT 1 TIME 11 (of 28) ]
[ time 34 ] Run JOB 0 at PRIORITY 1 [ TICKS 7 ALLOT 1 TIME 10 (of 28) ]
[ time 35 ] Run JOB 0 at PRIORITY 1 [ TICKS 6 ALLOT 1 TIME 9 (of 28) ]
[ time 36 ] Run JOB 0 at PRIORITY 1 [ TICKS 5 ALLOT 1 TIME 8 (of 28) ]
[ time 37 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 7 (of 28) ]
[ time 38 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 6 (of 28) ]
[ time 39 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 5 (of 28) ]
[ time 40 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 4 (of 28) ]
[ time 41 ] Run JOB 0 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 3 (of 28) ]
[ time 42 ] Run JOB 2 at PRIORITY 1 [ TICKS 9 ALLOT 1 TIME 9 (of 25) ]
[ time 43 ] Run JOB 2 at PRIORITY 1 [ TICKS 8 ALLOT 1 TIME 8 (of 25) ]
[ time 44 ] Run JOB 2 at PRIORITY 1 [ TICKS 7 ALLOT 1 TIME 7 (of 25) ]
[ time 45 ] Run JOB 2 at PRIORITY 1 [ TICKS 6 ALLOT 1 TIME 6 (of 25) ]
[ time 46 ] Run JOB 2 at PRIORITY 1 [ TICKS 5 ALLOT 1 TIME 5 (of 25) ]
[ time 47 ] Run JOB 2 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 4 (of 25) ]
[ time 48 ] Run JOB 2 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 3 (of 25) ]
[ time 49 ] Run JOB 2 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 2 (of 25) ]
[ time 50 ] Run JOB 2 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 1 (of 25) ]
[ time 51 ] Run JOB 2 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 0 (of 25) ]
[ time 52 ] FINISHED JOB 2
[ time 52 ] Run JOB 0 at PRIORITY 0 [ TICKS 19 ALLOT 4 TIME 2 (of 28) ]
[ time 53 ] Run JOB 0 at PRIORITY 0 [ TICKS 18 ALLOT 4 TIME 1 (of 28) ]
[ time 54 ] Run JOB 0 at PRIORITY 0 [ TICKS 17 ALLOT 4 TIME 0 (of 28) ]
[ time 55 ] FINISHED JOB 0

Final statistics:
  Job  0: startTime   0 - response   0 - turnaround  55
  Job  1: startTime   0 - response   5 - turnaround   7
  Job  2: startTime   0 - response   7 - turnaround  52

  Avg  2: startTime n/a - response 4.00 - turnaround 38.00
```

<table>
<tr><td>Que3</td><td>How would you configure the scheduler parameters to behave just like a round-robin scheduler?</td></tr>
<tr><td></td><td>
<pre>
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 3 -Q 5,5,5 -A 1,1,1 -B 0 -m 30 -M 0 -s 3 -c
Here is the list of inputs:
OPTIONS jobs 3
OPTIONS queues 3
OPTIONS allotments for queue 2 is   1
OPTIONS quantum length for queue 2 is   5
OPTIONS allotments for queue 1 is   1
OPTIONS quantum length for queue 1 is   5
OPTIONS allotments for queue 0 is   1
OPTIONS quantum length for queue 0 is   5
OPTIONS boost 0
OPTIONS ioTime 5
OPTIONS stayAfterIO False
OPTIONS iobump False

For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq   : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job 0: startTime   0 - runTime   7 - ioFreq   0
  Job 1: startTime   0 - runTime 11 - ioFreq   0
  Job 2: startTime   0 - runTime 19 - ioFreq   0

Execution Trace:

[time 0] JOB BEGINS by JOB 0
[time 0] JOB BEGINS by JOB 1
[time 0] JOB BEGINS by JOB 2
</pre>
</td></tr>
</table>

```
[ time 0 ] Run JOB 0 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 6 (of 7) ]
[ time 1 ] Run JOB 0 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 5 (of 7) ]
[ time 2 ] Run JOB 0 at PRIORITY 2 [ TICKS 2 ALLOT 1 TIME 4 (of 7) ]
[ time 3 ] Run JOB 0 at PRIORITY 2 [ TICKS 1 ALLOT 1 TIME 3 (of 7) ]
[ time 4 ] Run JOB 0 at PRIORITY 2 [ TICKS 0 ALLOT 1 TIME 2 (of 7) ]
[ time 5 ] Run JOB 1 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 10 (of 11) ]
[ time 6 ] Run JOB 1 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 9 (of 11) ]
[ time 7 ] Run JOB 1 at PRIORITY 2 [ TICKS 2 ALLOT 1 TIME 8 (of 11) ]
[ time 8 ] Run JOB 1 at PRIORITY 2 [ TICKS 1 ALLOT 1 TIME 7 (of 11) ]
[ time 9 ] Run JOB 1 at PRIORITY 2 [ TICKS 0 ALLOT 1 TIME 6 (of 11) ]
[ time 10 ] Run JOB 2 at PRIORITY 2 [ TICKS 4 ALLOT 1 TIME 18 (of 19) ]
[ time 11 ] Run JOB 2 at PRIORITY 2 [ TICKS 3 ALLOT 1 TIME 17 (of 19) ]
[ time 12 ] Run JOB 2 at PRIORITY 2 [ TICKS 2 ALLOT 1 TIME 16 (of 19) ]
[ time 13 ] Run JOB 2 at PRIORITY 2 [ TICKS 1 ALLOT 1 TIME 15 (of 19) ]
[ time 14 ] Run JOB 2 at PRIORITY 2 [ TICKS 0 ALLOT 1 TIME 14 (of 19) ]
[ time 15 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 1 (of 7) ]
[ time 16 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 0 (of 7) ]
[ time 17 ] FINISHED JOB 0
[ time 17 ] Run JOB 1 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 5 (of 11) ]
[ time 18 ] Run JOB 1 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 4 (of 11) ]
[ time 19 ] Run JOB 1 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 3 (of 11) ]
[ time 20 ] Run JOB 1 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 2 (of 11) ]
[ time 21 ] Run JOB 1 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 1 (of 11) ]
[ time 22 ] Run JOB 2 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 13 (of 19) ]
[ time 23 ] Run JOB 2 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 12 (of 19) ]
[ time 24 ] Run JOB 2 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 11 (of 19) ]
[ time 25 ] Run JOB 2 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 10 (of 19) ]
[ time 26 ] Run JOB 2 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 9 (of 19) ]
[ time 27 ] Run JOB 1 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 0 (of 11) ]
[ time 28 ] FINISHED JOB 1
[ time 28 ] Run JOB 2 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 8 (of 19) ]
[ time 29 ] Run JOB 2 at PRIORITY 0 [ TICKS 3 ALLOT 1 TIME 7 (of 19) ]
[ time 30 ] Run JOB 2 at PRIORITY 0 [ TICKS 2 ALLOT 1 TIME 6 (of 19) ]
```

```
[ time 31 ] Run JOB 2 at PRIORITY 0 [ TICKS 1 ALLOT 1 TIME 5 (of 19) ]
[ time 32 ] Run JOB 2 at PRIORITY 0 [ TICKS 0 ALLOT 1 TIME 4 (of 19) ]
[ time 33 ] Run JOB 2 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 3 (of 19) ]
[ time 34 ] Run JOB 2 at PRIORITY 0 [ TICKS 3 ALLOT 1 TIME 2 (of 19) ]
[ time 35 ] Run JOB 2 at PRIORITY 0 [ TICKS 2 ALLOT 1 TIME 1 (of 19) ]
[ time 36 ] Run JOB 2 at PRIORITY 0 [ TICKS 1 ALLOT 1 TIME 0 (of 19) ]
[ time 37 ] FINISHED JOB 2

Final statistics:
  Job  0: startTime   0 - response   0 - turnaround  17
  Job  1: startTime   0 - response   5 - turnaround  28
  Job  2: startTime   0 - response  10 - turnaround  37

  Avg  2: startTime n/a - response 5.00 - turnaround 27.33
```

Que4	Craft a workload with two jobs and scheduler parameters so that one job takes advantage of the older Rules 4a and 4b (turned on with the -S flag) to game the scheduler and obtain 99% of the CPU over a particular time interval.

```
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 2 -Q 5,10 -A 1,2 -B 0 -S -m 30 -M 0 -s 4 -c
Here is the list of inputs:
OPTIONS jobs 2
OPTIONS queues 2
OPTIONS allotments for queue  1 is   1
OPTIONS quantum length for queue  1 is   5
OPTIONS allotments for queue  0 is   2
OPTIONS quantum length for queue  0 is  10
OPTIONS boost 0
OPTIONS ioTime 5
OPTIONS stayAfterIO True
OPTIONS iobump False


For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq    : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job  0: startTime   0 - runTime   7 - ioFreq   0
  Job  1: startTime   0 - runTime  12 - ioFreq   0


Execution Trace:

[ time 0 ] JOB BEGINS by JOB 0
[ time 0 ] JOB BEGINS by JOB 1
[ time 0 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 6 (of 7) ]
[ time 1 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 5 (of 7) ]
[ time 2 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 4 (of 7) ]
[ time 3 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 3 (of 7) ]
```

```
[ time 4 ] Run JOB 0 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 2 (of 7) ]
[ time 5 ] Run JOB 1 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 11 (of 12) ]
[ time 6 ] Run JOB 1 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 10 (of 12) ]
[ time 7 ] Run JOB 1 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 9 (of 12) ]
[ time 8 ] Run JOB 1 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 8 (of 12) ]
[ time 9 ] Run JOB 1 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 7 (of 12) ]
[ time 10 ] Run JOB 0 at PRIORITY 0 [ TICKS 9 ALLOT 2 TIME 1 (of 7) ]
[ time 11 ] Run JOB 0 at PRIORITY 0 [ TICKS 8 ALLOT 2 TIME 0 (of 7) ]
[ time 12 ] FINISHED JOB 0
[ time 12 ] Run JOB 1 at PRIORITY 0 [ TICKS 9 ALLOT 2 TIME 6 (of 12) ]
[ time 13 ] Run JOB 1 at PRIORITY 0 [ TICKS 8 ALLOT 2 TIME 5 (of 12) ]
[ time 14 ] Run JOB 1 at PRIORITY 0 [ TICKS 7 ALLOT 2 TIME 4 (of 12) ]
[ time 15 ] Run JOB 1 at PRIORITY 0 [ TICKS 6 ALLOT 2 TIME 3 (of 12) ]
[ time 16 ] Run JOB 1 at PRIORITY 0 [ TICKS 5 ALLOT 2 TIME 2 (of 12) ]
[ time 17 ] Run JOB 1 at PRIORITY 0 [ TICKS 4 ALLOT 2 TIME 1 (of 12) ]
[ time 18 ] Run JOB 1 at PRIORITY 0 [ TICKS 3 ALLOT 2 TIME 0 (of 12) ]
[ time 19 ] FINISHED JOB 1

Final statistics:
  Job  0: startTime   0 - response   0 - turnaround  12
  Job  1: startTime   0 - response   5 - turnaround  19

  Avg  1: startTime n/a - response 2.50 - turnaround 15.50
```

Que5	Given a system with a quantum length of 10 ms in its highest queue, how often would you have to boost jobs back to the highest priority level (with the -B flag) in order to guarantee that a single longrunning (and potentially-starving) job gets at least 5% of the CPU?

```
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 2 -Q 5,10 -A 1,2 -B 10 -m 20 -M 0 -s 5 -c
Here is the list of inputs:
OPTIONS jobs 2
OPTIONS queues 2
OPTIONS allotments for queue  1 is   1
OPTIONS quantum length for queue  1 is   5
OPTIONS allotments for queue  0 is   2
OPTIONS quantum length for queue  0 is  10
OPTIONS boost 10
OPTIONS ioTime 5
OPTIONS stayAfterIO False
OPTIONS iobump False


For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq    : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job  0: startTime   0 - runTime  12 - ioFreq   0
  Job  1: startTime   0 - runTime  16 - ioFreq   0


Execution Trace:

[ time 0 ] JOB BEGINS by JOB 0
[ time 0 ] JOB BEGINS by JOB 1
[ time 0 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 11 (of 12) ]
[ time 1 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 10 (of 12) ]
[ time 2 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 9 (of 12) ]
[ time 3 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 8 (of 12) ]
```

	`` [time 4] Run JOB 0 at PRIORITY 1 [TICKS 0 ALLOT 1 TIME 7 (of 12)] [time 5] Run JOB 1 at PRIORITY 1 [TICKS 4 ALLOT 1 TIME 15 (of 16)] [time 6] Run JOB 1 at PRIORITY 1 [TICKS 3 ALLOT 1 TIME 14 (of 16)] [time 7] Run JOB 1 at PRIORITY 1 [TICKS 2 ALLOT 1 TIME 13 (of 16)] [time 8] Run JOB 1 at PRIORITY 1 [TICKS 1 ALLOT 1 TIME 12 (of 16)] [time 9] Run JOB 1 at PRIORITY 1 [TICKS 0 ALLOT 1 TIME 11 (of 16)] [time 10] BOOST (every 10) [time 10] Run JOB 0 at PRIORITY 1 [TICKS 4 ALLOT 1 TIME 6 (of 12)] [time 11] Run JOB 0 at PRIORITY 1 [TICKS 3 ALLOT 1 TIME 5 (of 12)] [time 12] Run JOB 0 at PRIORITY 1 [TICKS 2 ALLOT 1 TIME 4 (of 12)] [time 13] Run JOB 0 at PRIORITY 1 [TICKS 1 ALLOT 1 TIME 3 (of 12)] [time 14] Run JOB 0 at PRIORITY 1 [TICKS 0 ALLOT 1 TIME 2 (of 12)] [time 15] Run JOB 1 at PRIORITY 1 [TICKS 4 ALLOT 1 TIME 10 (of 16)] [time 16] Run JOB 1 at PRIORITY 1 [TICKS 3 ALLOT 1 TIME 9 (of 16)] [time 17] Run JOB 1 at PRIORITY 1 [TICKS 2 ALLOT 1 TIME 8 (of 16)] [time 18] Run JOB 1 at PRIORITY 1 [TICKS 1 ALLOT 1 TIME 7 (of 16)] [time 19] Run JOB 1 at PRIORITY 1 [TICKS 0 ALLOT 1 TIME 6 (of 16)] [time 20] BOOST (every 10) [time 20] Run JOB 0 at PRIORITY 1 [TICKS 4 ALLOT 1 TIME 1 (of 12)] [time 21] Run JOB 0 at PRIORITY 1 [TICKS 3 ALLOT 1 TIME 0 (of 12)] [time 22] FINISHED JOB 0 [time 22] Run JOB 1 at PRIORITY 1 [TICKS 4 ALLOT 1 TIME 5 (of 16)] [time 23] Run JOB 1 at PRIORITY 1 [TICKS 3 ALLOT 1 TIME 4 (of 16)] [time 24] Run JOB 1 at PRIORITY 1 [TICKS 2 ALLOT 1 TIME 3 (of 16)] [time 25] Run JOB 1 at PRIORITY 1 [TICKS 1 ALLOT 1 TIME 2 (of 16)] [time 26] Run JOB 1 at PRIORITY 1 [TICKS 0 ALLOT 1 TIME 1 (of 16)] [time 27] Run JOB 1 at PRIORITY 0 [TICKS 9 ALLOT 2 TIME 0 (of 16)] [time 28] FINISHED JOB 1 Final statistics: Job 0: startTime 0 - response 0 - turnaround 22 Job 1: startTime 0 - response 5 - turnaround 28 Avg 1: startTime n/a - response 2.50 - turnaround 25.00 ```

Que6	One question that arises in scheduling is which end of a queue to add a job that just finished I/O; the -I flag changes this behavior for this scheduling simulator. Play around with some workloads and see if you can see the effect of this flag.
	Run with -I IO RUN LATER

```
jitanjali_patel@jitanjali:~/os$ python3 7.py -j 2 -Q 5,10 -A 1,2 -B 0 -I IO RUN LATER -m 30 -M 5 -s 6
Here is the list of inputs:
OPTIONS jobs 2
OPTIONS queues 2
OPTIONS allotments for queue  1 is   1
OPTIONS quantum length for queue  1 is   5
OPTIONS allotments for queue  0 is   2
OPTIONS quantum length for queue  0 is  10
OPTIONS boost 0
OPTIONS ioTime 5
OPTIONS stayAfterIO False
OPTIONS iobump True


For each job, three defining characteristics are given:
  startTime : at what time does the job enter the system
  runTime   : the total CPU time needed by the job to finish
  ioFreq    : every ioFreq time units, the job issues an I/O
              (the I/O takes ioTime units to complete)

Job List:
  Job  0: startTime   0 - runTime  24 - ioFreq   4
  Job  1: startTime   0 - runTime  15 - ioFreq   2


Execution Trace:

[ time 0 ] JOB BEGINS by JOB 0
[ time 0 ] JOB BEGINS by JOB 1
[ time 0 ] Run JOB 0 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 23 (of 24) ]
[ time 1 ] Run JOB 0 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 22 (of 24) ]
[ time 2 ] Run JOB 0 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 21 (of 24) ]
[ time 3 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 20 (of 24) ]

[ time 3 ] Run JOB 0 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 20 (of 24) ]
[ time 4 ] IO_START by JOB 0
IO DONE
[ time 4 ] Run JOB 1 at PRIORITY 1 [ TICKS 4 ALLOT 1 TIME 14 (of 15) ]
[ time 5 ] Run JOB 1 at PRIORITY 1 [ TICKS 3 ALLOT 1 TIME 13 (of 15) ]
[ time 6 ] IO_START by JOB 1
IO DONE
[ time 6 ] IDLE
[ time 7 ] IDLE
[ time 8 ] IDLE
[ time 9 ] IO_DONE by JOB 0
[ time 9 ] Run JOB 0 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 19 (of 24) ]
[ time 10 ] Run JOB 0 at PRIORITY 0 [ TICKS 9 ALLOT 2 TIME 18 (of 24) ]
[ time 11 ] IO_DONE by JOB 1
[ time 11 ] Run JOB 1 at PRIORITY 1 [ TICKS 2 ALLOT 1 TIME 12 (of 15) ]
[ time 12 ] Run JOB 1 at PRIORITY 1 [ TICKS 1 ALLOT 1 TIME 11 (of 15) ]
[ time 13 ] IO_START by JOB 1
IO DONE
[ time 13 ] Run JOB 0 at PRIORITY 0 [ TICKS 8 ALLOT 2 TIME 17 (of 24) ]
[ time 14 ] Run JOB 0 at PRIORITY 0 [ TICKS 7 ALLOT 2 TIME 16 (of 24) ]
[ time 15 ] IO_START by JOB 0
IO DONE
[ time 15 ] IDLE
[ time 16 ] IDLE
[ time 17 ] IDLE
[ time 18 ] IO_DONE by JOB 1
[ time 18 ] Run JOB 1 at PRIORITY 1 [ TICKS 0 ALLOT 1 TIME 10 (of 15) ]
[ time 19 ] Run JOB 1 at PRIORITY 0 [ TICKS 9 ALLOT 2 TIME 9 (of 15) ]
[ time 20 ] IO_START by JOB 1
IO DONE
[ time 20 ] IO_DONE by JOB 0
[ time 20 ] Run JOB 0 at PRIORITY 0 [ TICKS 6 ALLOT 2 TIME 15 (of 24) ]
[ time 21 ] Run JOB 0 at PRIORITY 0 [ TICKS 5 ALLOT 2 TIME 14 (of 24) ]
[ time 22 ] Run JOB 0 at PRIORITY 0 [ TICKS 4 ALLOT 2 TIME 13 (of 24) ]
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[ time 23 ] Run JOB 0 at PRIORITY 0 [ TICKS 3 ALLOT 2 TIME 12 (of 24) ]
[ time 24 ] IO_START by JOB 0
IO DONE
[ time 24 ] IDLE
[ time 25 ] IO_DONE by JOB 1
[ time 25 ] Run JOB 1 at PRIORITY 0 [ TICKS 8 ALLOT 2 TIME 8 (of 15) ]
[ time 26 ] Run JOB 1 at PRIORITY 0 [ TICKS 7 ALLOT 2 TIME 7 (of 15) ]
[ time 27 ] IO_START by JOB 1
IO DONE
[ time 27 ] IDLE
[ time 28 ] IDLE
[ time 29 ] IO_DONE by JOB 0
[ time 29 ] Run JOB 0 at PRIORITY 0 [ TICKS 2 ALLOT 2 TIME 11 (of 24) ]
[ time 30 ] Run JOB 0 at PRIORITY 0 [ TICKS 1 ALLOT 2 TIME 10 (of 24) ]
[ time 31 ] Run JOB 0 at PRIORITY 0 [ TICKS 0 ALLOT 2 TIME 9 (of 24) ]
[ time 32 ] IO_DONE by JOB 1
[ time 32 ] Run JOB 1 at PRIORITY 0 [ TICKS 6 ALLOT 2 TIME 6 (of 15) ]
[ time 33 ] Run JOB 1 at PRIORITY 0 [ TICKS 5 ALLOT 2 TIME 5 (of 15) ]
[ time 34 ] IO_START by JOB 1
IO DONE
[ time 34 ] Run JOB 0 at PRIORITY 0 [ TICKS 9 ALLOT 1 TIME 8 (of 24) ]
[ time 35 ] IO_START by JOB 0
IO DONE
[ time 35 ] IDLE
[ time 36 ] IDLE
[ time 37 ] IDLE
[ time 38 ] IDLE
[ time 39 ] IO_DONE by JOB 1
[ time 39 ] Run JOB 1 at PRIORITY 0 [ TICKS 4 ALLOT 2 TIME 4 (of 15) ]
[ time 40 ] IO_DONE by JOB 0
[ time 40 ] Run JOB 0 at PRIORITY 0 [ TICKS 8 ALLOT 1 TIME 7 (of 24) ]

[ time 41 ] Run JOB 0 at PRIORITY 0 [ TICKS 7 ALLOT 1 TIME 6 (of 24) ]
[ time 42 ] Run JOB 0 at PRIORITY 0 [ TICKS 6 ALLOT 1 TIME 5 (of 24) ]
[ time 43 ] Run JOB 0 at PRIORITY 0 [ TICKS 5 ALLOT 1 TIME 4 (of 24) ]
[ time 44 ] IO_START by JOB 0
IO DONE
[ time 44 ] Run JOB 1 at PRIORITY 0 [ TICKS 3 ALLOT 2 TIME 3 (of 15) ]
[ time 45 ] IO_START by JOB 1
IO DONE
[ time 45 ] IDLE
[ time 46 ] IDLE
[ time 47 ] IDLE
[ time 48 ] IDLE
[ time 49 ] IO_DONE by JOB 0
[ time 49 ] Run JOB 0 at PRIORITY 0 [ TICKS 4 ALLOT 1 TIME 3 (of 24) ]
[ time 50 ] IO_DONE by JOB 1
[ time 50 ] Run JOB 1 at PRIORITY 0 [ TICKS 2 ALLOT 2 TIME 2 (of 15) ]
[ time 51 ] Run JOB 1 at PRIORITY 0 [ TICKS 1 ALLOT 2 TIME 1 (of 15) ]
[ time 52 ] IO_START by JOB 1
IO DONE
[ time 52 ] Run JOB 0 at PRIORITY 0 [ TICKS 3 ALLOT 1 TIME 2 (of 24) ]
[ time 53 ] Run JOB 0 at PRIORITY 0 [ TICKS 2 ALLOT 1 TIME 1 (of 24) ]
[ time 54 ] Run JOB 0 at PRIORITY 0 [ TICKS 1 ALLOT 1 TIME 0 (of 24) ]
[ time 55 ] FINISHED JOB 0
[ time 55 ] IDLE
[ time 56 ] IDLE
[ time 57 ] IO_DONE by JOB 1
[ time 57 ] Run JOB 1 at PRIORITY 0 [ TICKS 0 ALLOT 2 TIME 0 (of 15) ]
[ time 58 ] FINISHED JOB 1

Final statistics:
  Job  0: startTime   0 - response   0 - turnaround  55
  Job  1: startTime   0 - response   4 - turnaround  58

  Avg  1: startTime n/a - response 2.00 - turnaround 56.50
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